

Isaias Perez

isaiasxl21@gmail.com | [linkedin.com/in/isaias-perez21](https://www.linkedin.com/in/isaias-perez21) | <https://isaias-perez-portfolio.onrender.com> | (720) 980-7488

SUMMARY

Computer Science student with internships in AI/ML, software engineering, and quantitative research. Deployed ML models with **25% faster inference** (Azure/Databricks), optimized SQL backends for **12% faster queries**, and engineered a C++ trading simulator achieving **10x throughput** and **7% less slippage**. Proficient in C++, Python, React/Node.js, and cloud platforms (AWS, Azure, GCP). Lead Quant Developer for CU Quant Club, mentoring peers and directing strategy development. Active member of SHPE, driving collaborative projects, outreach, and contributing to diverse engineering initiatives.

EDUCATION

University of Colorado Boulder

Bachelor of Science in Computer Science

Major GPA: 3.78

Expected May 2027

Minors: Applied Mathematics, Business | **Courses:** Data Structures & Algorithms, Artificial Intelligence, Databases, Software Engineering, Applied Machine Learning, Probability, Statistics, Linear Algebra, Differential Equations, Finance, Accounting.

Affiliations: CU Quant Club (Lead Quant Developer, mentored members, directed strategy sessions, and built trading systems), Society of Hispanic Professional Engineers (SHPE, engaged in mentorship, outreach, and professional development)

TECHNICAL SKILLS

Languages: C++, Python, C#, JavaScript, SQL, HTML/CSS

Frameworks & OS: React, React Native, Node.js, Flask, Express.js, Git, Linux, Windows

ML & Data Tools: PyTorch, TensorFlow, Scikit-learn, NumPy, pandas, ONNX, XGBoost, LangChain, MLflow, MLOps, CI/CD

Platforms & Cloud: Microsoft Azure, AWS, Google Cloud, Databricks, BigQuery, Google Colab, Jupyter Notebooks, Docker

Other: Product Deployment, Team Leadership, Cross-functional Collaboration, Bilingual (English/Spanish)

EXPERIENCE

AI Specialist Intern

July 2025 – Present

Kroenke Sports & Entertainment (KSE)

Denver, CO

- Deployed ML models on Azure/Databricks, reducing inference latency by **25%** across **5+ AI services**.
- Fine-tuned LLM copilots on Azure ML, boosting reliability and cutting latency for a **30% workflow gain**.
- Analyzed **100K+ firewall logs** with PySpark, automating **4+ compliance checks** and reducing review by **35%**.

AI Engineer Intern

June 2025 – Present

Polylabs

Remote

- Built *Upwordly*, an AI LinkedIn generator fine-tuned on **1M+ tokens**, raising engagement by **22%**.
- Designed multimodal pipelines (OpenAI, Hugging Face, n8n) automating **100+ posts/month** and cutting workload **40%**.

Machine Learning Research Assistant

March 2025 – Present

University of Colorado Boulder

Boulder, CO

- Collaborated on ICML paper, leading experiments and statistical analysis of complex synthetic datasets.
- Engineered attribution/saliency tools to study **4+ transformers**, lowering hallucinations by **15%**.

Software Engineer Intern

Apr 2023 – Aug 2023

Ryno Industries LLC

Erie, CO

- Optimized SQL queries and backend modules, cutting spec latency by **12%** for **3+ sales teams**.
- Redesigned dashboard (React + Bootstrap), boosting conversions **18%** across **200+ monthly sessions**.

PROJECTS

FlowExec++ | Reinforcement Learning for Trade Execution

C++, PyTorch, RL, Gemini API

- Engineered a **high-performance trade execution simulator** in C++ with Heston stochastic volatility and market impact models; achieved **10x** faster simulation throughput after optimization and quantization.
- Trained and tuned PPO and Flow Matching agents for low-latency execution, reducing slippage by **7%** on LOBSTER.

PulsePlan (AI Scheduler) | Multimodal LLM Agent for Task Automation

React Native, Node.js, OpenAI, Gemini

- Built AI productivity planner using GPT-4 and Gemini for calendar, tasks, and habit tracking.
- Created web scraping engine for LMS integration (Canvas, Moodle) and deployed Chrome extension with Stripe billing.

EcoVision | Edge-AI for Smart Waste Sorting

Python, YOLOv8, ONNX, FastAPI

- Engineered real-time waste classification system using YOLOv8 and custom labeled dataset.
- Quantized and exported models to ONNX for NPU-accelerated inference on low-power edge devices.